

Diandian Peng

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Education

Ph.D. in Geology, University of Illinois at Urbana-Champaign (UIUC) May 2022

Thesis: *Quantifying slab evolution and mantle flow using global subduction models with data assimilation.*

Advisor: Lijun Liu. Committee: Jay Bass, Patricia Gregg, and William Guenther.

M.S. in Geophysics, University of Science and Technology of China (USTC) June 2017

Thesis: *Geodynamic modeling of the evolution around the southeastern Tibetan Plateau.*

B.S. in Geophysics, School of the Gifted Young, USTC June 2014

Thesis: *A preliminary dynamical simulation for the evolution of the Tibetan Plateau.*

Research Experience

Independent Researcher 2025 – present

- Stepped away from a full-time academic appointment to be a full-time parent, while staying active in the research community.
- Continue to review manuscripts for journals in geodynamics, tectonics, and mantle dynamics, and welcome new review invitations.

Postdoctoral Scholar, Scripps Institution of Oceanography, UC San Diego 2022 – 2025

- Modeled the Tonga–Kermadec subduction zone, the fastest-retreating trench on Earth, with mantle-convection simulations.
- Investigated the formation of Large Low-Shear-Velocity Provinces (LLSVPs) and mantle plumes.
- Studied plume–slab interactions and their geological implications.

Graduate Research Assistant, UIUC 2017 – 2022

- Developed global and regional mantle-convection models with data assimilation.
- Investigated the formation of unusual slab geometries and their surface expression.
- Studied seismic anisotropy and the migration of slab material with mantle flow.

Graduate Research Assistant, USTC 2014 – 2017

- Studied the uplift of the Tibetan Plateau and its steep margins.
- Analyzed crustal channel flow using analytical and numerical methods.

Undergraduate Research Assistant, USTC 2013 – 2014

- Explored the evolution of the Tibetan Plateau with numerical models.
- Studied the thermal structure of the continental lithosphere from heat-flow data.

Research Interests

- Developing and applying novel numerical-modeling techniques.
- The interplay between mantle dynamics and tectonic plates.
- The diverse range of subduction processes and their mechanisms.
- The formation, evolution, and dynamics of LLSVPs and mantle plumes.
- Plume–slab interactions and the resulting lithospheric deformation and volcanism.
- Using geophysical, geochemical, and geological data to constrain mantle dynamics and lithospheric structure.

Computational & Technical Skills

- **Geodynamic modeling:** global and regional mantle-convection modeling with sequential data assimilation; forward and adjoint (retrodiction) approaches linking deep-mantle flow to surface observables.
- **Modeling & visualization tools:** CitcomS, ASPECT (mantle convection); GPlates (plate reconstructions); ParaView, VisIt, GMT.
- **Programming:** Python (NumPy, SciPy, Pandas, Matplotlib), C, C++, Fortran, MATLAB, Bash; Git, Linux.
- **Methods & computing:** finite-element and finite-difference methods, parallel / high-performance computing (TACC Frontera), inverse problems, signal processing, and machine learning.

Publications

Peer-reviewed journal articles

16. Zhu, T., **Peng, D.** and Liu, L. (2025). High-speed anomaly zone beneath the Changbaishan volcanic province in NE China: Insights from simulation of the Cenozoic Pacific plate subduction using geodynamic models with data assimilation. *Journal of Asian Earth Sciences*. [link]
15. Zhu, T., **Peng, D.** and Liu, L. (2025). Factors affecting the Pacific plate subduction towards and under the Changbaishan volcanic province since the Cenozoic: Insights from geodynamic modeling based on data assimilation. *Tectonophysics*. [pdf]
14. Zhu, T., **Peng, D.** and Liu, L. (2025). Effects of 410- and 660-km seismic discontinuities on the Pacific plate subduction and slab geometry beneath the Changbaishan volcanic province. *Acta Seismologica Sinica* (in Chinese).
13. **Peng, D.** and Stegman, D.R. (2024). Modeling subduction with extremely fast trench retreat. *Journal of Geophysical Research: Solid Earth*. [pdf]
12. **Peng, D.** and Stegman, D.R. (2024). Geodynamic evolution of the Lau Basin. *Geophysical Research Letters*. [pdf]
11. Li, Y.^{*}, Liu, L., Li, S., **Peng, D.**, Cao, Z. and Li, X. (2024). Cenozoic India-Asia collision driven by mantle dragging the cratonic root. *Nature Communications*. [pdf]
10. Xue, T., **Peng, D.**, Liu, K.H., Obrist-Farner, J., Locmelis, M., Gao, S.S. and Liu, L. (2023). Ongoing fragmentation of the subducting Cocos slab, Central America. *Geology*. [pdf]
9. Li, Y.^{*}, Liu, L., **Peng, D.**, Dong, H. and Li, S. (2023). Evaluating tomotectonic plate reconstructions using geodynamic models with data assimilation, the case for North America. *Earth-Science Reviews*. [pdf]
8. Wang, Y., Cao, Z., Peng, L., Liu, L., Chen, L., Lundstrom, C., **Peng, D.** and Yang, X. (2023). Secular craton evolution due to cyclic deformation of underlying dense mantle lithosphere. *Nature Geoscience*. [pdf]
7. **Peng, D.** and Liu, L. (2023). Importance of global spherical geometry for studying slab dynamics and evolution in models with data assimilation. *Earth-Science Reviews*. [pdf]
6. Liu, Y.^{*}, Liu, L., Li, Y.^{*}, **Peng, D.**, Wu, Z., Cao, Z., Li, S. and Du, Q. (2022). Global back-arc extension due to trench-parallel mid-ocean ridge subduction. *Earth and Planetary Science Letters*. [pdf]
5. **Peng, D.** and Liu, L. (2022). Quantifying slab sinking rates using global geodynamic models with data assimilation. *Earth-Science Reviews*. [pdf]
4. **Peng, D.**, Liu, L. and Wang, Y. (2021). A newly discovered Late-Cretaceous East Asian flat slab explains its unique lithospheric structure and tectonics. *Journal of Geophysical Research: Solid Earth*. [pdf]
3. **Peng, D.**, Liu, L., Hu, J., Li, S. and Liu, Y.^{*} (2021). Formation of East Asian stagnant slabs due to a pressure-driven Cenozoic mantle wind following Mesozoic subduction. *Geophysical Research Letters*. [pdf]
2. Liu, L., **Peng, D.**, Liu, L., Chen, L., Li, S., Wang, Y., Cao, Z. and Feng, M. (2021). East Asian lithospheric evolution dictated by multistage Mesozoic flat-slab subduction. *Earth-Science Reviews*. [pdf]
1. **Peng, D.** and Leng, W. (2017). Analytical and numerical simulations of uplift processes at the Tibet-Sichuan boundary. *Earthquake Science*. [pdf]

Book

1. Li, S., Zhu, J., Cao, X., Liu, L., Zhou, J. and **Peng, D.** (2023). *Atlas of Global Mantle Microplates with Tomographic Images*. Science Press (Beijing). ISBN 9787030752086. Selected as one of the "Outstanding Marine Science and Technology Books for 2024" by the Chinese Society for Oceanography.

Preprint

1. Suo, Y., Dong, H., Liu, L., **Peng, D.**, Li, Y. *, Liu, J., Dai, L., Cao, X. and Li, S. (2022). Landward mantle flow associated with the Pacific subduction system opened the South China Sea.

In preparation

2. **Peng, D.**, Stegman, D.R., Day, J.M.D., Parnell-Turner, R., Liu, L. and Wang, Y. Intraplate Indian Ocean magmatism and the Toba supervolcano produced by the Capricorn plume.
1. **Peng, D.**, Liu, L., Cao, Z., Li, Y. *, Stegman, D.R. and Wang, Y. Cenozoic uplift of the western US due to removal of dense lithospheric roots.

* Student I mentored.

Grants & Computing Allocations

TACC Frontera — Simulating slab–plume interaction with realistic mantle convection models (PI: Dave Stegman) 2023 – 2025

Awarded 224,640 SUs (12.6 M core-hours) in 2025, 230,520 SUs (12.9 M) in 2024, and 225,000 SUs (12.6 M) in 2023. Contributed to the scientific justification, usage plan, and proposal, and ran the computations on Frontera.

NSF EAR 2244660 — Investigating formation of stagnant slabs and implications for subduction dynamics (\$333,141; PIs: Lijun Liu and Craig Lundstrom) 2023
Contributed to the scientific justification and figures.

NSF OAC 2031704 — NSF Frontera Allocation Travel Grant (\$10,000; PI: Lijun Liu) 2020
Contributed to the scientific justification and figures.

Honors & Awards

Cecil H. and Ida M. Green Postdoctoral Fellowship (IGPP, Scripps, UC San Diego) — \$138,000 2022 – 2024

Bluestem Fellowship (UIUC) 2017 – 2018

Graduate Student First-Class Academic Scholarship (USTC) 2014 – 2016

Outstanding Student Paper Award (CGU) 2015

National Encouragement Scholarship (USTC) 2013

Outstanding Student Scholarship (USTC) 2013

Outstanding Freshman Scholarship (USTC) 2010

Invited Talks

University of South Florida — "Where the mantle wind blows." 2025

Nanjing University — "Applications of mantle convection models across diverse subduction zones, and insights into overriding-plate deformation." 2024

Southern University of Science and Technology (SUSTech) — "Studying subduction-zone dynamics using numerical models of different dimensions." 2024

Chinese University of Hong Kong — "Studying circum-Pacific subduction zones with geodynamic models." 2023

Scripps Institution of Oceanography, UC San Diego — IGPP Seminar Series. 2023

Conference Presentations

- Liu, L., Cao, Z., Li, Y.^{*}, Li, X., Dong, H. and **Peng, D.** (2026). Revisiting the temporal evolution of oceanic subduction. *EGU General Assembly*.
- Liu, L., **Peng, D.** and Li, Y.^{*} (2023). What drives uplift of west-central North America since the Late Cretaceous? *AGU Fall Meeting*.
- Liu, L., Li, Y.^{*}, **Peng, D.** and He, X. (2023). Investigating East Asian lithospheric evolution since the Cretaceous. *AGU Fall Meeting*.
- Li, Y.^{*}, Liu, L. and **Peng, D.** (2022). What drives the post-collisional northward Indian motion? *AGU Fall Meeting*.
- Peng, D.**, Liu, L., Chen, L. and Li, S. (2021). East Asian lithospheric evolution in response to west Pacific subduction since 100 Ma. *AGU Fall Meeting*.
- Peng, D.** and Liu, L. (2021). Quantifying horizontal vs. vertical motions of subducted slabs using global geodynamic models with data assimilation. *AGU Fall Meeting*.
- Liu, L., **Peng, D.**, Liu, L., Li, Y.^{*} and Chen, L. (2021). A new geodynamic framework for the evolution from intraplate orogeny to continental extension. *AGU Fall Meeting*.
- Li, Y.^{*}, **Peng, D.**, Cao, Z. and Liu, L. (2021). Continental-scale converging mantle flow beneath Eurasia and America, from model to observation. *AGU Fall Meeting*.
- Liu, L., Zhou, Q. and **Peng, D.** (2021). Cenozoic topographic evolution of western-central US. *GSA Annual Meeting*.
- Peng, D.**, Liu, L., Chen, L. and Li, S. (2020). Late-Cretaceous Izanagi flat subduction below East Asia and tectonic responses. *AGU Fall Meeting*.
- Liu, L., **Peng, D.**, Wang, Y. and Hu, J. (2020). Differences and similarities of lithosphere deformation during flat-slab subduction and plume underplating. *AGU Fall Meeting*.
- Peng, D.**, Liu, L., Liu, Y.^{*}, Hu, J. and Li, S. (2019). Formation of East Asian stagnant slabs due to a westward Cenozoic mantle wind. *AGU Fall Meeting*.
- Liu, L., Hu, J., Zhou, Q., Liu, Y. and **Peng, D.** (2019). Formation of slab tears controls Earth's major silicic volcanism. *AGU Fall Meeting*.
- Peng, D.**, Liu, L. and Hu, J. (2018). Understanding subduction dynamics in the Southwest Pacific. *AGU Fall Meeting*.
- Peng, D.** and Leng, W. (2016). Analytical and numerical simulation of uplift at the Tibet–Sichuan boundary. *AGU Fall Meeting*.

* Student I mentored.

Teaching & Mentoring

Research mentoring, UIUC

- Yiming Liu (visiting graduate student) — guided research led to a publication in *Earth and Planetary Science Letters*.
- Yanchong Li (graduate student) — guided research led to publications in *Earth-Science Reviews* and *Nature Communications*, plus several AGU presentations.

Teaching assistant, UIUC

- GEOL 104 — Geology of the National Parks (Spring 2021).
- GEOL 117 — The Oceans (Fall 2018, Fall 2020).

Teaching assistant, USTC

- Introduction to Earth Science (Fall 2016).

Fieldwork

R/V *Thomas G. Thompson* research cruise, South Pacific Ocean Nov – Dec 2023

- Installed and deployed ocean-bottom seismometers (OBS) near the Lau Basin and the Samoa Hotspot to collect seismic data.
- Conducted dredging near the Samoa Hotspot and the Cook Islands, and cut, classified, and described the recovered rock samples on board.

Baraboo, Wisconsin	2021
Kentland and Williamsport, Indiana	2019
Sugar Creek, Indiana	2017
Chaohu, Anhui, China	2014

Service

- **Manuscript reviewer** for *Geophysical Research Letters*, *Journal of Geophysical Research: Solid Earth*, *Geology*, *Tectonophysics*, *Geochemistry*, *Geophysics*, *Geosystems* (G-Cubed), *Physics of the Earth and Planetary Interiors*, *Geoscience Letters*, *Proceedings of the Royal Society A*, *PNAS Nexus*, and *Scientific Reports*.
- Department colloquium coordinator, UIUC (Fall 2018).
- Volunteer, Engineering Open House, UIUC.

Outreach

- "Why do some slabs stagnate?" — EGU Geodynamics Blog (2022). [\[link\]](#)